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PAPER TITLE

**Quantum-Evolved Motion Priors in Particle-Based
Missile Guidance**

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The Problem

High-G Maneuvers

Fixed PN models cannot anticipate curvature. Estimation error compounds through the guidance loop until interception fails.

ECM & Decoys

Spectrally matched decoys and electronic jamming distort measurement updates and sever datalink mid-flight.

Estimation Drift

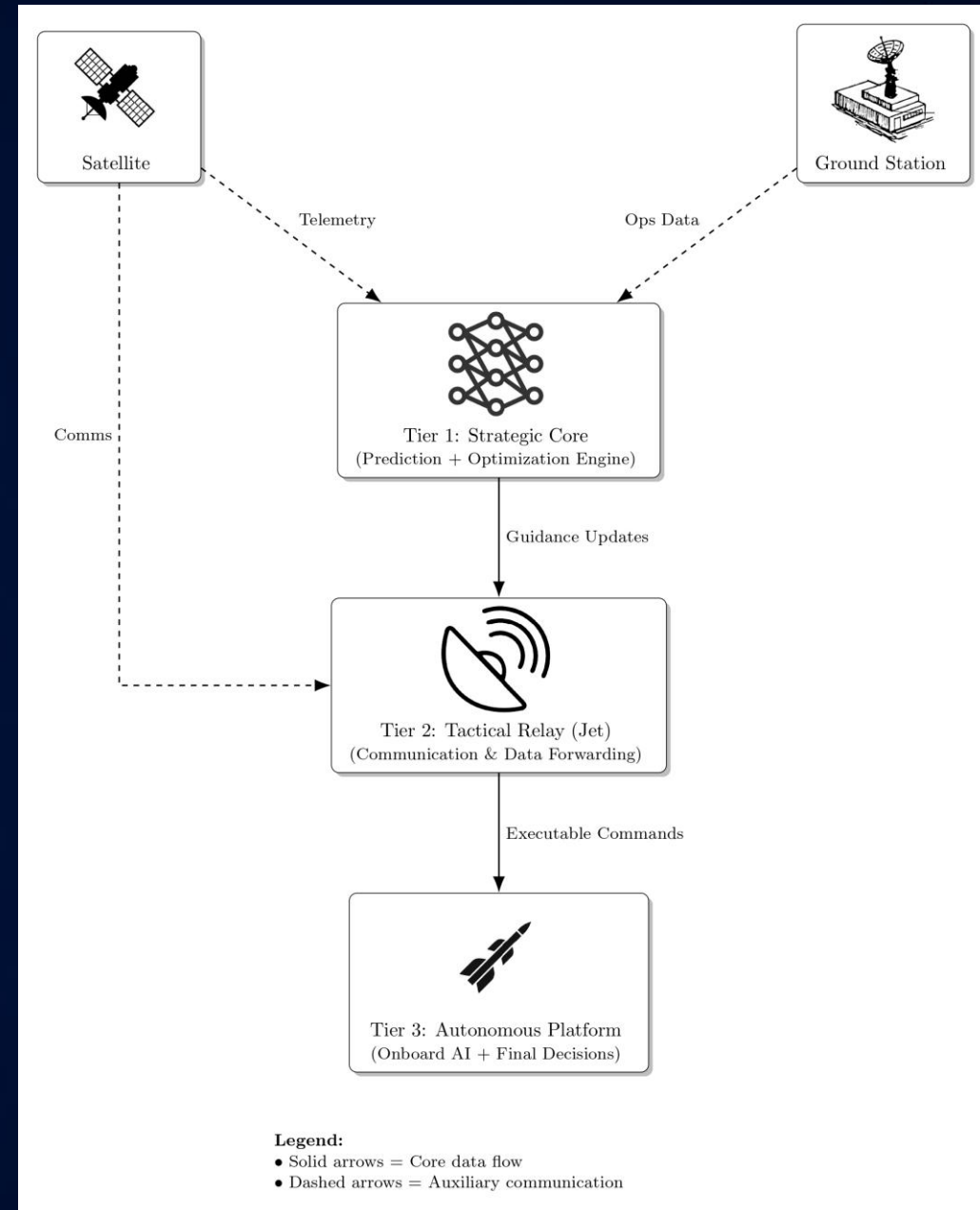
EKF/UKF linearization error accumulates exponentially under aggressive motion. Particle filters help but still use memoryless priors.

The gap is not estimation accuracy — it is the absence of prediction-aware propagation that adapts structurally to maneuver geometry.

Proposed Architecture

- **Tier 1 – Strategic Core (Offboard)**
 - PF + QNEAT
 - Predictive Guidance
 - Trajectory Optimization
- **Tier 2 – Tactical Relay**
 - Communication Bridge
 - ECM Handling
- **Tier 3 – Autonomous Missile**
 - Local Filtering
 - PN Fallback

Graceful Degradation:
Predictive Guidance → Reduced Updates → Autonomous PN



QNEAT: How It Works?

Inputs

- Particle state x_k
- Kinematic features ϕ_k (angular rate, closing geometry)

QNEAT Evolution (Offline)

- Evolves discrete quantum circuit topologies
- Gate set: {H, X, CNOT}, 4 qubits, 256 shots
- No gradient descent
- No large-scale labelled dataset
- Fitness rewards both acceleration & velocity prediction

Output

- Motion increment Δx_k replaces fixed kinematic step
 - $x_{k+1}^{(i)} = x_k^{(i)} + \Delta x_k^{(i)} + w_k^{(i)}$
- Next State
=
Current State + QNEAT Prediction + Uncertainty

What Changes

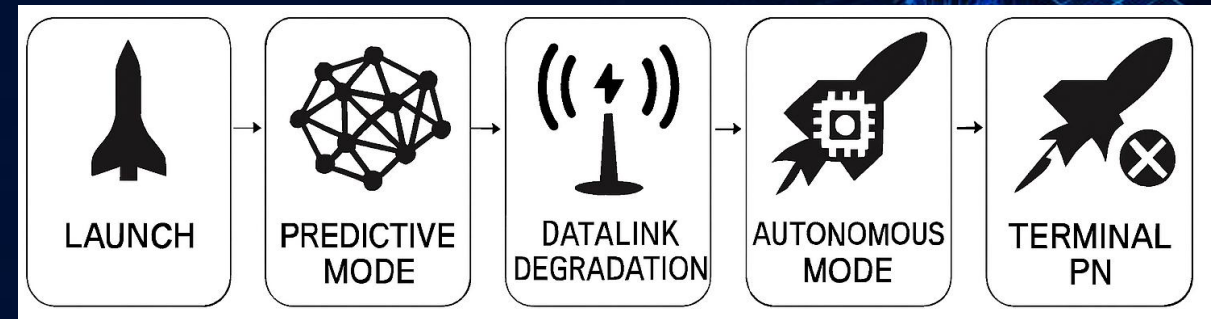
- Particles are propagated toward maneuver-consistent futures
- Bayesian weights remain untouched
- No quantum hardware required — classical simulation

Guidance Optimization & Mode Transitions

Trajectory Re-Optimization Layer

- Executed offboard within Tier 1.
- Guidance optimization formulated as a discrete trajectory selection problem.
- QAOA incorporated as the proposed optimization framework.
- Generates guidance advisories for mid-course updates.
- No quantum computation onboard the interceptor.

Graceful Mode Transitions



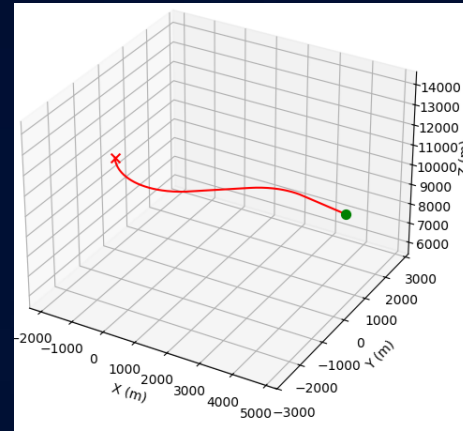
Simulation Setup



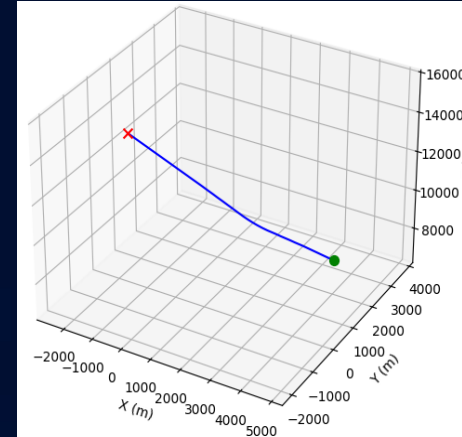
Parameters Table

Parameter	Value
Particle count (N)	500
Qubits (nq)	4
Measurement shots (S)	256
Evolution generations	200
Population size (P)	100
Fitness weight (α)	0.5
Branching horizon (H)	3 steps
Integration timestep	0.05 s
Simulation mode	Classical (no quantum HW)

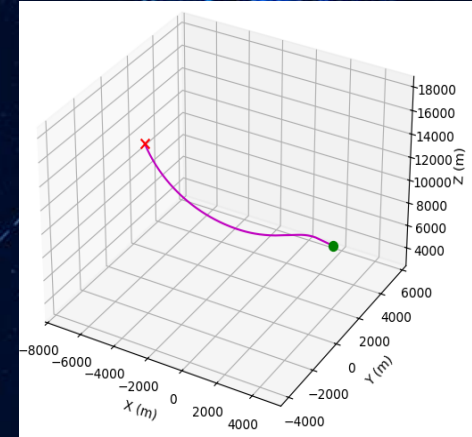
Maneuver Profiles



Profile 1:
Oscillatory + Vertical



Profile 2:
Bidirectional Turn



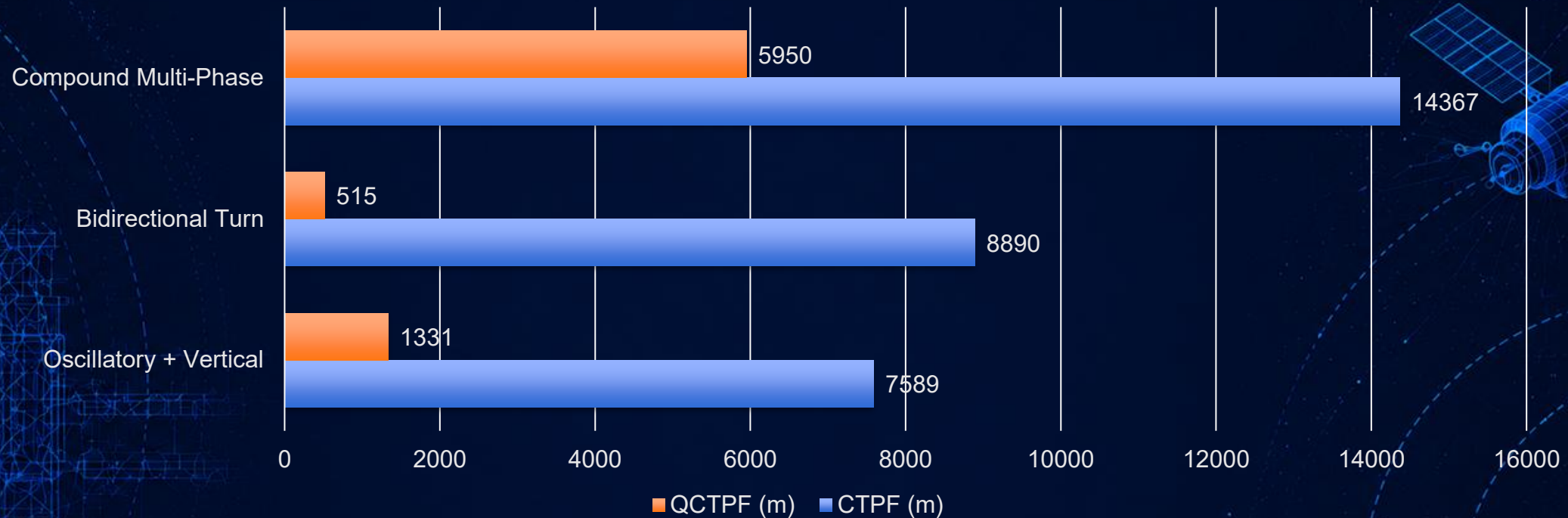
Profile 3:
Compound Multi-Phase

Only the prediction mechanism differs; all sensing, guidance, and vehicle dynamics remain identical.

Interception Performance: Terminal Miss Distance



Terminal Miss Distance (m) — Lower is Better



Improved maneuver alignment translates into significantly lower terminal miss distance.

82% ↓

Oscillatory +
Vertical

94% ↓

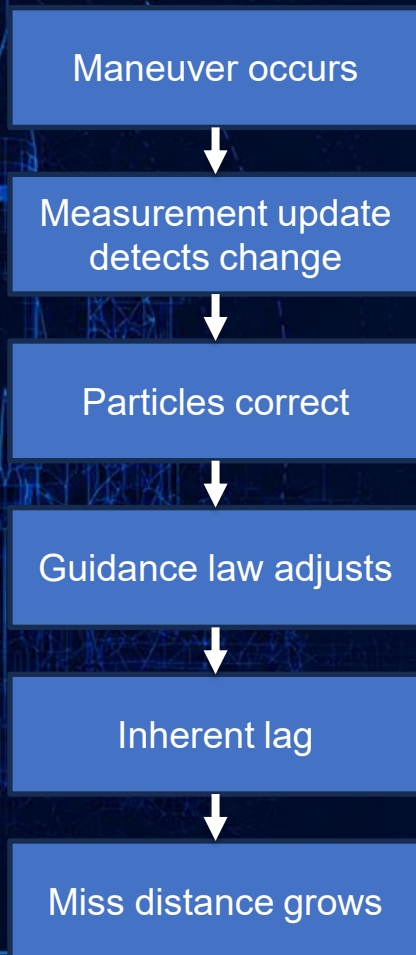
Bidirectional
Turn

59% ↓

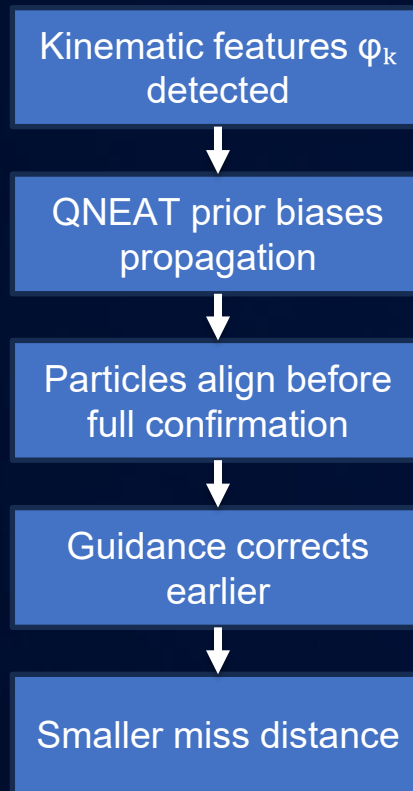
Compound
Multi-Phase

Why QCTPF Improves Interception Performance?

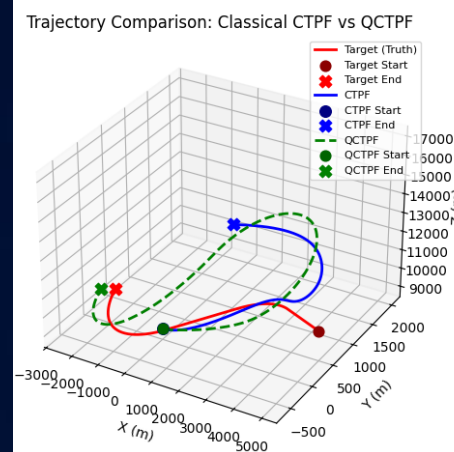
Classical PF (reactive)



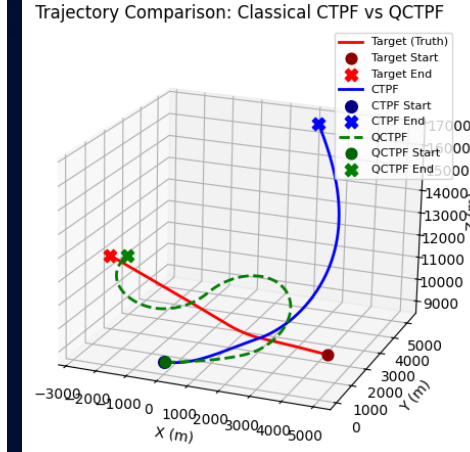
QCTPF (predictive)



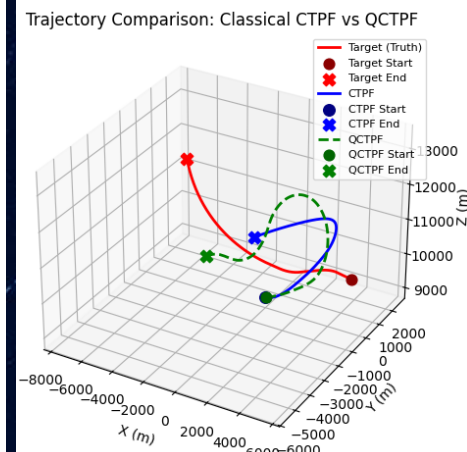
*Bayesian update remains unchanged;
only particle propagation is modified.*



Profile 1:
Oscillatory + Vertical



Profile 2:
Bidirectional Turn



Profile 3:
Compound Multi-Phase

Novelty & Key Contributions

Prior Work vs Our Distinction

Prior Work	Limitation	Our Distinction
PN-GAPF	Prior unchanged	Motion prior itself is learned
Wang et al.	Deterministic optimization	Prediction-aware propagation
RL-based guidance	Dataset dependence	Evolutionary learning, no large dataset
Distributed estimation	No predictive layer	Integrated relay + prediction architecture

Key Contributions

QNEAT Motion Prior

Replaces fixed kinematic particle propagation

Predictive Guidance Framework

Generates maneuver-aware mid-course advisories

Three-Tier Resilient Architecture

Prediction → Relay → Autonomous PN

Experimental Validation

- 3 nonlinear maneuver profiles evaluated
- Up to 94% reduction in terminal miss distance

Conclusion

- ✓ Maneuver-aware particle propagation reduces reactive lag during target maneuvers.
- ✓ Earlier trajectory alignment produces significantly lower terminal miss distance.
- ✓ Resilient operation is maintained through graceful degradation under link disruption.

Up to 94% reduction in terminal miss distance

Thank You

Questions & Discussion

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